TBdecompile Reference Manual

Appendix C — TBX File Format

Schema and analysis of actual exports

15 October 2026

1. What TBX is

TBX is the XML exchange format used by ToolBook 9.5 and 11.5. It is not the on-disk TBK format. It is the expanded, human-readable form produced by File → Save To XML when saveSpecial=true is set in Instructor115.ini.

A TBKX package is a ZIP of a TBX folder. The folder contains Book.xml, one Page.xml per page, Background.xml files, .script files, and resource files (bitmaps, etc.).

Namespace: urn:sumtotalsystems:toolbook.tbkx

2. Schema hierarchy (from Denny / TBSME)

```
Object
  +- ToolBookObject (IdNumber, Name, SharedScript, Script)
       +- GraphicObject (RgbFill, RgbStroke, Transparent, Visible)
            +- DrawObject (Bounds required, LineStyle, Pattern)
            |    +- VectorObject (Vertices optional)
            |         +- LineObject (LineEndSize, LineEndStyle)
            +- TextObject (Bounds, FontFace, FontSize, FontStyle)
                 +- FieldClass (Text, BorderStyle, FieldType)
```

Bounds is required on every DrawObject. Pattern: -?\d{1,5},-?\d{1,5},-?\d{1,5},-?\d{1,5} (left,top,right,bottom in twips).

Vertices is optional on VectorObject. Bounds-only is valid. The tip shape is expanded at paint or export time.

Schema typo: IrregularPloygon (missing 'y'). Official exports use this spelling.

3. Book structure

```
Book
  +- Book.Pages          PageLocator list (IdNumber, Source path)
  +- Book.Backgrounds    BackgroundLocator list
  +- Book.Resources      Bitmap, Cursor, Font, Icon, Menubar,
  |                      SharedScript, Clip
  +- Book.Viewers
  +- Book.UserProperties
  +- Book.Actions
  Size default: 8640,5760 (6" x 4" at 1440 twips/inch)
```

From the 9e954d01 export:

```
Book Script="Book.script" Caption="Nida Corporation - Logic Families"
     Size="9360,6624" MinorVersionNumber="664"
     xmlns="urn:sumtotalsystems:toolbook.tbkx"
```

4. Page structure

```
Page
  +- Page.Objects        sequence of Image, Arc, Button, Curve,
  |                      Ellipse, Field, Group, IrregularPloygon,
  |                      Line, PaintObject, Picture, Pie, Polygon,
  |                      Rectangle, RoundedRectangle, Stage, ...
  +- Page.UserProperties
  +- Page.Actions
  Attributes: Name, IdNumber (required), Background (required),
              Script, SkipNavigation, SharedScript, DefaultAllowDrop
```

The 9e954d01 export contains 131 Page.xml files. Two of these (P440, P2920) have no corresponding page record in the raw TBK.

5. Object examples from 9e954d01

P1040 (Page880) — TTL intro, two DIP packages:

```
IrregularPolygon  28   pins, all with Vertices
Rectangle          2   bodies (fill 64,64,64)
Arc                2
Field              5
Button             3
Group              5   left (5400) and right (7400) packages
```

First pin Vertices (twips):

```
Vertices="5055,4605,5205,4605,5205,4733,5175,4767,5160,4825,
          5145,4860,5115,4860,5100,4825,5085,4767,5055,4733,
          5055,4698"
```

Binary record at the same location stores only the bounding box:

```
5055, 4605, 5205, 4860   then   00 00 00 00 c0 c0 c0
```

The numbers match at the start. The tip is computed, not stored. TBX expands it.

6. P430 and P420

P430 (Page875) — chip diagram:

96 IrregularPolygons with Vertices. 8 Groups, 8 Rectangles, 8 Arcs. Matches the live GDI!POLYGON traces (11-point notches). The binary only exposed a fraction of these as explicit point records.

P420 — flame:

IrregularPolygons with very long Vertices lists (one dark-red, one brighter red). These are the flame curves that rendered as a tangle when point order was taken from the raw record. Also 18 AngledLines.

7. Scripts

Book.script contains the enterSystem handler that pushes tb80r.sbk, nc81base.sbk, nc81test.sbk, nc81recs.sbk, nc81trnr.sbk and calls the initialize* / openSession functions.

Per-object .script files are plain OpenScript:

```
to handle buttonClick
  go to page "P400"
end
```

In the TBK these are compressed bytecode streams. In TBX they are already expanded.

8. Resources

Bitmaps are external files (Bitmap100.bmp, ...). Bitmap resources have KeyColor and UseChromaKey for transparency. SharedScript is a resource, not inline. No embedded OLE or other complex resources were found in the 9e954d01 export.

Backgrounds are first-class objects with their own XML and .script files.

9. TBX vs TBK

```
TBX attribute          Binary (9e954d01.tbk)
--------------         ---------------------
Bounds="l,t,r,b"       four-word record + fill color
Vertices="x,y,..."     kind=0400 point list at +0x22
                       (or bounds-only; tip expanded)
RgbFill="r,g,b"        fill color dword at +4
Script="..."           bytecode in page blob
IdNumber               page / object id
Background             [?] not a direct word in the page record
```

TBX is the expanded form. The binary stores bounds; TBX expands Vertices. A clean-room renderer can use Vertices/Bounds directly instead of reverse-engineering every 14 00 00 layout.

10. Counts from 9e954d01

```
Pages:              131 Page.xml
Backgrounds:        multiple BackgroundN.xml
IrregularPolygons:  654 total (across all pages)
Scripts:            Book.script + per-object .script files
Bitmaps:            external Bitmap*.bmp
```

The large number of IrregularPolygons explains why earlier binary walks under-counted shapes. Many pins are bounds-only in the TBK and only become explicit point lists in the TBX.

11. Implications for a clean-room reader

TBX is a convenient source for accurate geometry. The Vertices and Bounds attributes can be used directly. Scripts can be used as-is; no bytecode decoding is needed for this lesson.

For a reader that must also handle raw TBK:

- Use TBX as the reference for what the expanded form should look like.
- Treat bounds-only binary records as incomplete; the tip is computed.
- Groups in TBX show which objects form a visual unit.
- The two extra pages (P440, P2920) in TBX have no binary counterpart; they may be generated or unused.